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$$\det(a) := a$$
$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} := ad - bc$$
$$\det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} := a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - c_1 b_2 a_3 - c_2 b_3 a_1 - c_3 b_1 a_2.$$

Alternative notation for determinant: $|\mathbf{A}| = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$. (This is a scalar, not a matrix!)

Why we care about determinants.

The inverse of a matrix **A** exists if and only if $\det \mathbf{A} \neq 0$.

How to compute determinants.

One way to compute determinants is with the aid of a computer algebra system like MATLAB. However, there are methods for computing the determinant of any matrix by hand as well.

Laplace expansion (along the first row) to compute the determinant of a **3 × 3** matrix:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = +a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}.$$

The general rule leading to the formula above is this:

1. Move your finger along the entries in a row.
2. At each position, compute the **minor** , defined as the smaller determinant obtained by crossing out the row and the column through your finger; then multiply the minor by the number you are pointing at, and adjust the sign according to the checkerboard pattern

$$\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

(the pattern always starts with + in the upper left corner).

3. Add up the results.

There is a similar expansion for a determinant of any size, computed along any row or column.

Practice the Laplace expansion formula

1/1 point (graded)
Suppose that

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Calculate **detA**. Enter your answer as a number (both fractional and decimal are okay).

detA =  Answer: 0

Solution:

Using Laplace expansion along the first row we see that

$$\begin{aligned} \det A &= 1 \cdot \det \begin{pmatrix} 5 & 6 \\ 8 & 9 \end{pmatrix} - 2 \cdot \det \begin{pmatrix} 4 & 6 \\ 7 & 9 \end{pmatrix} + 3 \cdot \det \begin{pmatrix} 4 & 5 \\ 7 & 8 \end{pmatrix} \\ &= -3 - 2(-6) + 3(-3) = 0. \end{aligned}$$

Determinants and transpose matrices

1/1 point (graded)
The **transpose** of a matrix **A** is another matrix **A^T** whose columns are the rows of **A**. For example, the matrix

$$\mathbf{B} = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

is the transpose of the matrix **A** of the previous problem.

Calculate **detB**. Enter your answer as a number (both fractional and decimal are okay).

detB =

 Answer: 0

Solution:

The Laplace expansion along the first **column** of **B** is the same as the Laplace expansion along the first **row** of **A** (from the previous problem), so **detB = detA = 0**.
The same observation about the transpose of any square matrix shows that

$$\det \mathbf{A}^T = \det \mathbf{A}.$$

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13. Determinants

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